
Convincing Biology with Language: Offline Learning of Prompt-Conditioned Biological Interventions

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Abstract

Artificial intelligence has transformed how people interact with complex technical systems across domains – from writing code to designing molecules – enabling non-experts to accomplish sophisticated tasks through natural language prompts. We envision an analogous future for biotechnology: scientists and practitioners directing living matter by describing what they want, with learned models translating those descriptions into the interventions that convince cells, organoids, or engineered biological systems to behave accordingly. Learning this mapping requires paired language-intervention-outcome data, which is expensive to collect at scale. Recent work has demonstrated language-conditioned policies for simulated cellular control, but these rely on hand-engineered reward functions and task-specific architectures, and do not exploit existing observational archives, requiring fresh experiments for every new objective. Here we demonstrate, for the first time, that a policy network can be trained offline, from a fixed archive of intervention-outcome pairs, to translate natural language prompts into interventions shown to reliably elicit desired behaviors in *in vitro* motile organoids (Xenobots). We frame this as offline contextual-bandit learning, with a three-stage pipeline: (1) a prompt-to-intervention (P2I) network maps language embeddings to intervention parameters, (2) a video archive lookup retrieves the historical Xenobot response to that intervention, and (3) a pretrained vision-language model provides a zero-shot reward signal scoring how well the observed behavior matches the original prompt. Optimizing the P2I network end-to-end with CMA-ES, and resolving objective interference between opposing behavioral objectives via a multi-head architecture, we demonstrate generalization to 40 unseen test prompts and 44 previously unobserved Xenobot videos, with 6 of 8 phenotypes exceeding the random-retrieval baseline. This work shows that language-guided biological control is learnable from purely observational data, without online experiments or hand-crafted rewards, and constitutes an intermediate step toward convincing biology with natural language.

1 Introduction

Controlling living systems is a long-standing goal across biology, medicine, and synthetic biology, yet the interface remains fundamentally low-level: practitioners specify interventions parameter-by-parameter, through trial-and-error informed by domain expertise. In other technical domains, natural language has emerged as a high-level interface to complex systems: users now specify goals — generate this code, design this molecule, plan this trajectory — and learned models translate intent into low-level action. This raises a question central to applied ML: can a similar interface be learned for biological systems, mapping natural-language descriptions of desired behavior to physical interventions that elicit it?

Two recent threads make this question tractable. Language-conditioned policies have been shown to control simulated cellular systems [? ?], and targeted physical interventions — electrical, optogenetic, chemical — have been shown to elicit reproducible behaviors in real cells and tissues [? ?]. However, extending this to real organisms under open-ended language faces two coupled barriers. First, collecting paired (prompt, intervention, outcome) data online is prohibitively expensive: each datum is a wet-lab experiment. Second, the objective space is inherently multi-task and contains opposing goals — *e.g.*, “stop” versus “accelerate” — which create objective interference when optimized through a shared output [? ?]. Existing approaches sidestep both barriers by restricting to hand-crafted reward functions or single-task architectures, and as a result do not generalize to open-ended natural language.

We address both barriers by reframing language-to-intervention as an offline learning problem. Existing biological archives already contain large numbers of intervention-outcome pairs collected for purposes other than language-conditioned learning; we treat these as a fixed dataset over which a policy can be trained without any new experiments. To supply a reward signal in the absence of labeled prompt-outcome alignment, we use a pretrained vision-language model (VLM) as a zero-shot scorer [?], evaluating how well an archived outcome matches a natural-language description. To address the second barrier, the policy uses a shared backbone with one output head per behavioral phenotype, enabling independent optimization across opposing objectives while preserving shared semantic representation [?].

Our contributions are: (1) A demonstration that the language-to-intervention mapping can be learned offline, from a fixed archive of intervention-outcome pairs collected without language annotation, eliminating the need for new wet-lab experiments per objective; (2) An end-to-end pipeline combining a prompt-to-intervention policy network, archive lookup, and VLM-based zero-shot rewards, optimized with CMA-ES; and (3) A multi-head policy architecture that resolves objective interference between opposing behavioral objectives, enabling a single network to fit antagonistic prompts (*e.g.* “stop” and “accelerate”) simultaneously.

2 Related Work

Large language models have demonstrated remarkable ability to serve as general-purpose interfaces to complex technical domains. Scaling has enabled few-shot task learning [?], with subsequent work showing LLMs can generate functional code [?], design molecules [?], and assist in scientific discovery [?]. The common pattern is that users specify goals through natural language rather than low-level parameters. However, applying LLMs to biological control has proven difficult. Most prior work in this direction restricts to simulation, where prompts map to cell parameters with deterministic outcomes [? ?]. Direct application to living organisms is rare, as the mapping from language to biological intervention lacks formal grounding — unlike code, biology offers no closed-form semantics for what move faster should mean as an electrical stimulus. Moreover, collecting paired data (prompt, intervention, outcome) requires expensive real-time experiments.

Synthetic biology has produced sophisticated engineered systems with autonomous behavior [? ?]. Xenobots, living robots assembled from *Xenopus laevis* cells, exhibit spontaneous coordinated motion and adaptive morphology [3]. Bioelectric networks encode high-level anatomical information, suggesting goal-directed control mechanisms [?]. Yet control of these systems remains manual and low-level: practitioners tune stimulation parameters (duration, current, frequency) through trial-and-error, without any high-level behavioral interface, and the intervention-to-behavior mapping is non-linear and poorly characterized. Targeted physical interventions are nonetheless effective at the parameter level: dynamic electric cues can steer cell migration with precision [?], and optogenetic stimulation can probe biological decision-making [?]. What is missing is a learned interface between high-level behavioral specifications and these low-level intervention protocols.

Bridging language and biological behavior requires two innovations. Vision-language models trained on joint image-text embeddings [? ?] can be used directly as zero-shot reward models, scoring the alignment between an observation and a language description [? ?]. This approach has been explored in robotics [? ?]; biological applications remain limited because the relevant behaviors are subtle, the visual backgrounds cluttered, and the outcomes stochastic. Learning a single policy across opposing behavioral goals (“stop” vs. “accelerate”) is a multi-task problem, and shared outputs

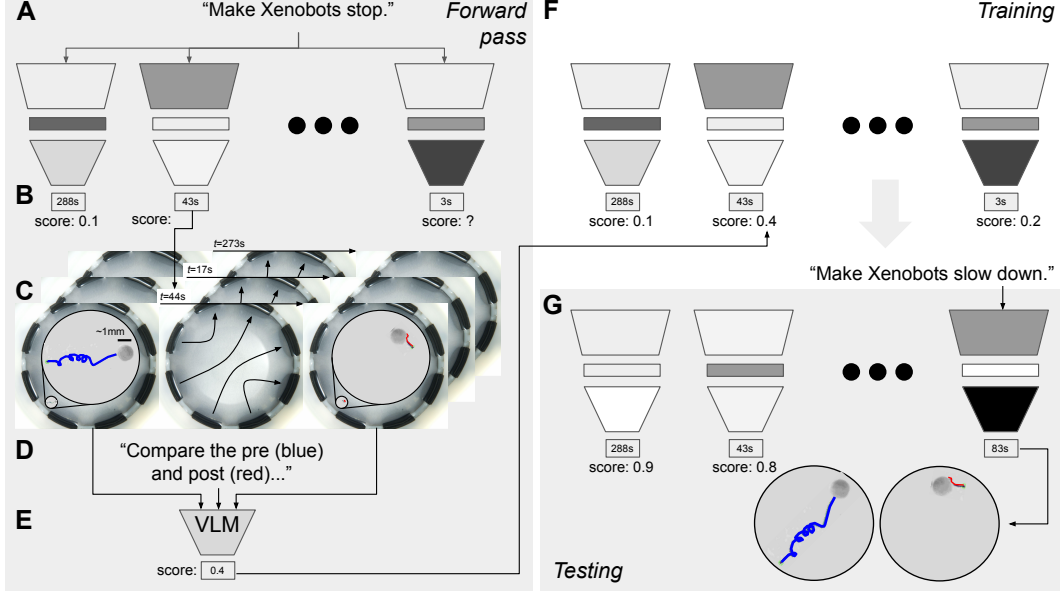


Figure 1: The pipeline. (A) User prompt $\rightarrow$ (B) P2I network $\rightarrow$ intervention duration $\rightarrow$ (C) archive lookup $\rightarrow$ (D) pre/post motion heatmaps $\rightarrow$ (E) VLM score. (F) CMA-ES evolves the P2I population on training prompts; (G) the trained P2I generalizes to held-out prompts and batches. Detailed in Sect. 3.

are known to suffer objective interference [7?]; task-specific heads over a shared backbone are a standard architectural remedy [? ?].

Finally, training from a fixed archive of intervention-outcome pairs places this work in the offline RL setting [?], and specifically in the contextual-bandit regime: there are no state transitions, only terminal rewards. This regime is well-suited to derivative-free black-box optimization, for which CMA-ES [?] is a standard choice — gradient-free, robust to multi-modal landscapes, and effective in the parameter dimensionalities considered here.

3 Methods

Our approach (Fig. 1) has two phases. In the forward pass, a natural-language prompt (Sect. 3.1) is mapped by a policy to an intervention (Sect. 3.2); the closest intervention is retrieved from an archive (Sect. 3.3), which has been split into training and test interventions (Sect. 3.4); the resulting video (Sect. 3.5) and prompt are scored by a VLM (Sect. 3.6). In the backward pass, evolutionary search optimizes the policy to maximize VLM scores on training prompts (Sect. 3.7).

3.1 Prompts.

Users specify desired behavior through natural language descriptions (e.g., “make Xenobots stop moving”, “increase activity”). Each prompt is embedded using a pretrained sentence transformer (SentenceBERT; [?]), producing a 384-dimensional embedding that captures semantic meaning. This embedding serves as the input to the P2I network.

3.2 Prompt-to-Intervention (P2I):

The policy network π_{Θ} maps prompt embeddings to behavior-conditioned intervention durations. Given prompt embedding $\mathbf{e} = \phi(p) \in \mathbb{R}^{384}$ (SentenceBERT), a shared convolutional backbone extracts features (detailed architecture and parameter counts in Appendix A):

$$\mathbf{f} = \text{GlobalAvgPool}(\text{Conv1D}_{384 \rightarrow 16}(\mathbf{e})) \quad (1)$$

where the three-layer CNN reduces dimensionality ($384 \rightarrow 64 \rightarrow 32 \rightarrow 16$) with ReLU activations and dropout (0.2).

115 We compare two output designs. The *single-head baseline* produces one scalar for all behaviors:

$$\mu_{\text{single}} = \sigma(\text{MLP}_{16 \rightarrow 1}(\mathbf{f}; \Theta_{\text{single}})) \quad (2)$$

116 Under this design, prompts that demand opposing interventions – “stop” (short stimulus) versus
 117 “accelerate” (long stimulus) – backpropagate conflicting signals through the same output, so the
 118 network is pulled toward incompatible targets. Our *multi-head design* decouples this by giving each
 119 of the 8 behaviors its own output head:

$$\mu_b(\mathbf{e}) = \sigma(\text{MLP}_{16 \rightarrow 32 \rightarrow 1}(\mathbf{f}; \Theta_b)), \quad b \in \{0, 1, \dots, 7\} \quad (3)$$

120 Independent heads allow specialization (each behavior optimizes its Θ_b independently) while the
 121 shared backbone (hard parameter sharing [?]) encodes semantic features useful across all behaviors.
 122 With a 7,984-parameter backbone and 577 parameters per behavior head, the 8-behavior multi-head
 123 architecture uses 12,600 total parameters versus 8,561 for single-head, a 47.2% increase that enables
 124 behavior specialization. During training, prompt-behavior pairs are routed to the correct head via
 125 oracle labels; at test time, routing uses cosine similarity to the nearest training-prompt embedding,
 126 enabling generalization to unseen phrasings.

127 3.3 The Archive.

128 Our dataset comprises 143 Xenobot intervention videos collected from an academic institution’s
 129 bioelectricity research center (between October 2025 and January 2026. Videos available upon
 130 request with institutional approval). Each video captures pre- and post-intervention behavior at 30
 131 fps, 1280×960 resolution. Xenobot responses depend on multiple intervention parameters (current
 132 magnitude, stimulation angle, frequency, duration); we fix current, angle, and frequency, and treat
 133 *stimulus duration* $d \in [1, 327]$ seconds (median: 131s) as the single intervention variable. Each batch
 134 consists of (1) raw video, (2) a command log with exact intervention parameters and timing, and
 135 (3) trajectory files. Restricting variation to a single parameter isolates the language-to-intervention
 136 mapping from confounds in the underlying biology.

137 3.4 Data Splitting

138 The 143-batch archive is stratified into 99 training and 44 test batches, balancing difficulty tiers within
 139 each split. This stratification exploits a natural property of the data: VLM baseline scores (computed
 140 on random batch selection) reveal a bimodal distribution. *Easy-tier* behaviors (stop, motion_reduction,
 141 elimination, confinement, response_magnitude) receive consistently high baseline scores averaging
 142 0.74, indicating the archive contains abundant examples of these phenotypes. *Hard-tier* behaviors
 143 (directedness, vigor, motion_increase) receive sparse, low baseline scores averaging 0.16, indicating
 144 these phenotypes are rare in the archive (Table 1). This distribution justifies the stratified train-test
 145 split and defines the difficulty landscape: behaviors that are naturally abundant require minimal
 146 prompting, while rare behaviors demand genuine language-conditioned learning to discover them.

147 We use 80 total prompts: 40 for training and 40 held-out for testing. The 40 training prompts (8
 148 behaviors × 5 prompts) are partitioned 3:2 per behavior into optimization and validation prompts.
 149 The 99 training batches support both optimization and validation; the 44 test batches remain held-out.

150 3.5 Videos.

151 Raw Xenobot videos (30 fps, 1280×960) contain optical noise from bubbles and electrode reflections.
 152 To extract clean behavioral signals, we denoise in three steps: (1) frame differencing to localize
 153 motion regions, (2) HSV saturation thresholding ($s > 20$) to reject low-saturation artifacts (bubbles,
 154 reflections), preserving pigmented Xenobot regions, and (3) morphological cleanup of the resulting
 155 mask (detailed in Appendix D).

156 Dense optical flow (Farneback method) computed over the denoised mask yields per-frame motion
 157 magnitude. We accumulate motion magnitude separately over pre-intervention and post-intervention
 158 periods, producing two motion heatmaps that visualize behavior change:

$$\Delta H = \sum_t \text{Magnitude}_{\text{masked}}^{\text{post}}(t) - \sum_t \text{Magnitude}_{\text{masked}}^{\text{pre}}(t) \quad (4)$$

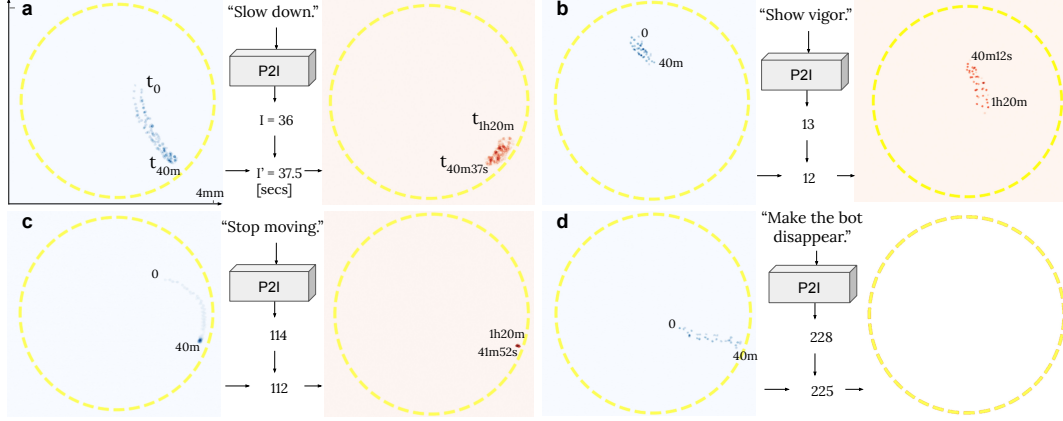


Figure 2: **Four examples of language-guided behavior produced by the trained P2I network.** For each prompt, the network outputs an intervention duration i , rounded to the nearest archived duration i' ; panels show the pre- (blue) and post-intervention (red) Xenobot trajectories in the matched archive batch. Yellow dashed circles denote the chamber boundary (4 mm diameter; 142 pixels/mm).

Figure 2 shows examples of pre- and post-intervention motion heatmaps computed from four distinct batches in the archive. These heatmaps serve as visual inputs to the VLM reward signal (Sect. 3.6), which evaluates alignment between the observed behavioral change and the command prompt given to the P2I network.

3.6 The VLM.

For each (video, prompt) pair, we compute an alignment score using a frozen vision-language model (Qwen3.5-397B served via Ollama Cloud). The VLM takes as input: (1) pre- and post-intervention motion heatmaps from Eq. 4, and (2) a query template populated with the command prompt given to the P2I network: “Compare pre (blue) vs post (red) heatmaps. How well does post-intervention behavior match: {prompt}? Score 0 - 1.” The VLM’s scalar output $s \in [0, 1]$ represents zero-shot alignment - how faithfully the observed behavior change matches the requested prompt - and serves as the reward signal. Because this score is deterministic per (video, prompt, behavior) triple under greedy decoding, we precompute the full score tensor $\mathbf{S} \in \mathbb{R}^{143 \times 80 \times 8}$ once, prior to optimization, eliminating redundant VLM calls during training. Detailed VLM query templates are provided in Appendix A.

3.7 Training

We optimize policy parameters θ to maximize average reward across all behaviors and their training prompts:

$$\max_{\theta} \frac{1}{8} \sum_{b=1}^8 \mathbb{E}_{\mathbf{e} \in P_b} [r(\theta, \mathbf{e}, b)] \quad (5)$$

where $b \in \{1, \dots, 8\}$ is the behavior index, P_b is the set of training prompts for behavior b , the action $a \in [0, 1]$ is the intervention duration produced by the policy on prompt embedding $\mathbf{e} \in \mathbb{R}^{384}$, and the reward $r \in [0, 1]$ is the VLM score (Sect. 3.6). Each behavior’s reward is averaged across its training prompts. This is a one-step contextual bandit: terminal rewards only, no state transitions, 143 samples per behavior [? ?]. We adopt CMA-ES [?], a derivative-free algorithm suited to sparse, non-differentiable rewards, with population size $\lambda = 4 + \lfloor 3 \ln(n) \rfloor = 32$ for $n = 12,600$ multi-head parameters, run for 100 generations with initial mutation strength $\sigma_0 = 0.3$.

Table 1: The eight behavioral phenotypes used in this work. Operational definitions characterize the post-intervention motion pattern that defines each behavior; example prompts illustrate the natural-language descriptions used at training and test time. Train/Test baselines report mean VLM scores under random archive matching; these represent the inherent average behavior score of the dataset, achieved by randomly sampling durations.

Behavior	Example prompt	Train	Test
<i>Decrease in motion</i>			
stop	“Stop moving.”	0.750	0.743
motion_reduction	“Move slower.”	0.763	0.810
<i>Increase in motion</i>			
motion_increase	“Move faster.”	0.140	0.135
vigor	“Show vigor.”	0.156	0.152
response_magnitude	“Dramatic behavior change.”	0.762	0.824
<i>Structure of motion</i>			
directedness	“Show purposeful movement.”	0.150	0.190
<i>Spatial extent</i>			
confinement	“Stay confined to center.”	0.631	0.654
elimination	“Bot ceases all activity.”	0.700	0.758

184 4 Results

185 4.1 Training

186 We compared CMA-ES variants (Table 2): single-head and multi-head architectures, against random
 187 archive sampling as a reference control. The random baseline samples uniformly random durations
 188 from the archive for each prompt; over many samples, this converges to the dataset’s inherent average
 189 behavior score (0.510 across all behaviors).

190 CMA-ES multi-head achieved 0.9269 ± 0.0096 training fitness and 0.7522 ± 0.0137 validation fitness,
 191 demonstrating strong learning and generalization. Single-head CMA-ES reached 0.6664 ± 0.0218
 192 training fitness but only 0.3999 ± 0.1124 validation fitness, indicating severe overfitting: the shared
 193 backbone struggles to balance opposing objectives, causing poor transfer to held-out data. Multi-head
 194 specialization enables strong training and generalization despite the additional parameters.

Table 2: Overall training and validation fitness across optimization methods (mean $\pm$ SD across 30 seeds). Baseline is the mean VLM score from the training dataset. Values in parentheses show improvement (+) or degradation (-) vs. baseline.

Method	Train Fitness	Val Fitness
CMA-ES (multi-head)	0.9269 ± 0.0096 (+81.5%)	0.7522 ± 0.0137 (+47.5%)
CMA-ES (single-head)	0.6664 ± 0.0218 (+30.6%)	0.3999 ± 0.1124 (-21.6%)
Baseline	0.510	0.510

195 4.2 Testing

196 The training validation results revealed that both single-head CMA-ES and REINFORCE fail to learn
 197 across the full behavioral repertoire: single-head suffers from severe overfitting (0.3999 validation
 198 fitness, -21.6% vs baseline), while REINFORCE achieves only marginal gains that do not transfer
 199 (0.426 validation fitness, -16.5% vs baseline). Although these methods may reach acceptable scores
 200 on individual easy behaviors, they cannot achieve a compromise solution balancing all 8 objectives.
 201 Accordingly, we focus generalization analysis on the successful approach: CMA-ES multi-head,

which demonstrated strong validation performance (0.7522, +47.5% vs baseline). Detailed per-behavior analysis of single-head and REINFORCE is provided in Appendix C.

We evaluate generalization across two protocols. **GT1** (Generalization to novel prompts) pairs 40 held-out test prompts with the 99 training-set videos, measuring transfer to unseen language. **GT2** (Generalization to novel batches) pairs the same 40 test prompts with 44 held-out videos from a different experimental cohort, testing transfer across both prompt and intervention space.

First, we compute overall average test scores across all 8 behaviors and 30 seeds (Table 3). Both conditions show highly significant improvement over the random baseline. GT1 achieves strong transfer to novel prompts (0.866 mean score, +70.9% improvement, $p < 10^{-41}$). GT2 shows substantial degradation when batches are also novel (0.627 mean score, +23.8% improvement, $p < 10^{-20}$), but remains significantly above baseline.

Table 3: Overall test generalization performance: averaged across all 8 behaviors and 30 seeds. Baseline is the mean VLM score from the dataset (average of all possible random intervention selections). GT1 demonstrates strong transfer to novel prompts; GT2 shows moderate transfer when both prompts and batches are novel.

Condition	Mean Score	Improvement	p -value
Baseline	0.510	—	—
GT1 (test prompts $\times$ train videos)	0.866	+70.9%	$p < 10^{-41}$
GT2 (test prompts $\times$ test videos)	0.627	+23.8%	$p < 10^{-20}$

Per-behavior results reveal distinct generalization patterns (Appendix B). Under GT1, all eight behaviors exceed $p < 10^{-7}$, with gains scaling inversely to baseline difficulty: the three hardest behaviors — `motion_increase` (baseline 0.140), `directedness` (0.150), and `vigor` (0.156) — achieve +425%, +388%, and +375% improvements respectively, reaching 0.73–0.74. Easy-tier behaviors near ceiling show smaller relative but substantial absolute gains, with `response_magnitude` reaching 0.980.

Under GT2, six behaviors maintain positive transfer. The opposing behaviors `vigor` (+307%), `motion_increase` (+186%), and `directedness` (+128%) remain substantially improved. However, `confinement` and `motion_reduction` show negative transfer (-53.1% and -26.9% respectively), likely reflecting test-cohort duration distribution differences that affect these phenotypes’ sensitivity to stimulus duration. We return to this point in Sect. 5. Further details, including per-behavior statistics (Cohen’s d , R^2) and inter-behavior correlations, are provided in Appendix B.

5 Discussion

5.1 What we learned

Language-guided biological control is learnable from offline observational data without custom reward functions. The key architectural insight is that multi-task learning with multi-head decoupling and shared backbone substantially outperforms isolated single-task specialists (Appendix A.2): This pattern aligns with [?] — shared representations enable hard tasks to benefit from easier ones, preventing overfitting. The network demonstrates robust semantic understanding, successfully generalizing to novel language inputs.

5.2 Limitations

Data scale and scope. The archive is small (143 batches: 99 training, 44 test), potentially missing rare intervention patterns and limiting the range of duration-outcome relationships the network can learn. Evaluation covers only 8 behaviors in Xenobots; transfer to other organoid systems or whole organisms remains untested. The network selects from existing interventions rather than generating novel ones, restricting it to the parameter space explored during data collection.

Reward signal validity. The VLM reward is unvalidated against human expert raters, risking systematic bias or artifacts in the scoring pipeline. This is particularly concerning for subjective phenotypes (`directedness`, `vigor`) where human and algorithm disagreement could significantly skew results.

Population specificity and the GT1-GT2 gap. The sharp drop from GT1 (71.7%) to GT2 (20.8%) reveals a fundamental limitation: the network learns a population-specific mapping from prompt to intervention to phenotype. GT1 succeeds because test prompts pair with the same population of Xenobots. GT2 tests distribution shift: test videos come from a different biological population with distinct baseline activity, growth stage, and responsiveness. The network captures language semantics but not the underlying biological principles governing how interventions affect different populations.

5.3 Future work

Our immediate priorities are human validation of VLM scoring, and active data collection targeting high-uncertainty regions of intervention space. Closing the loop with real-time experiments enables feedback-driven refinement. Architectural extensions include generative models to synthesize novel interventions, and lifelong learning for incremental behavior discovery. Transfer experiments to other biological systems would reveal whether language-to-intervention learning generalizes across cell types and scales.

Broader Impact

This work translates natural-language descriptions into electrical interventions for Xenobots. The method lowers the barrier for researchers without electrophysiology expertise. A key risk is that non-experts may not recognize when an intervention has unintended consequences – for example, a request to "make the bot disappear" retrieved an archived experiment where prolonged stimulation possibly destroyed the Xenobot. Since the system retrieves only archived interventions, this risk is bounded. Broader ethical questions about Xenobot research apply here as well.

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339 Appendix

340 Code and Data Availability

341 Code, data, and analysis are provided on GitHub: [https://github.com/drartificiallife/](https://github.com/drartificiallife/Neurips2026-Xenobots)
342 Neurips2026-Xenobots. Upon publication acceptance, the Xenobots experimental data will be
343 released according to institutional data policies and collaborator agreements.

344 Appendix A: P2I Network Architecture Details

345 .1 Shared Backbone

346 Input: 384-dimensional SentenceBERT embedding

347 Convolutional layers: Three 1D convolutions reducing dimensionality

$$\text{Conv1D}(384 \rightarrow 64, k = 3) \rightarrow \text{ReLU} \quad (6)$$

$$\text{Conv1D}(64 \rightarrow 32, k = 3) \rightarrow \text{ReLU} \quad (7)$$

$$\text{Conv1D}(32 \rightarrow 16, k = 3) \rightarrow \text{ReLU} \quad (8)$$

348 Global average pooling: Output is $\mathbf{f}_{\text{shared}} \in \mathbb{R}^{16}$

349 .2 Per-Behavior Heads

350 Each of 8 behavior heads shares the backbone but has independent downstream layers:

$$\mu_b = \sigma(\text{FC}_{32 \rightarrow 1}(\text{ReLU}(\text{FC}_{16 \rightarrow 32}(\mathbf{f}_{\text{shared}}))))), \quad b \in \{0, 1, \dots, 7\} \quad (9)$$

351 where σ is sigmoid (output range $[0, 1]$ for intervention duration scaling).

352 .3 Parameter Count

Backbone (shared Conv1d layers):

$$\text{Conv1d}(384 \rightarrow 64, k = 3) : 384 \times 64 \times 3 + 64 = 73,792$$

$$\text{Conv1d}(64 \rightarrow 32, k = 3) : 64 \times 32 \times 3 + 32 = 6,176$$

$$\text{Conv1d}(32 \rightarrow 16, k = 3) : 32 \times 16 \times 3 + 16 = 1,552$$

$$\text{Backbone Total} : 73,792 + 6,176 + 1,552 = \mathbf{81,520} \text{ parameters}$$

Output heads (per behavior):

$$\text{Linear}(16 \rightarrow 32) : 16 \times 32 + 32 = 544$$

$$\text{Linear}(32 \rightarrow 1) : 32 \times 1 + 1 = 33$$

$$\text{Per-head Total} : 544 + 33 = \mathbf{577} \text{ parameters}$$

Network totals:

$$\text{Single-head network} : 7,984 + 577 = 8,561 \text{ parameters}$$

$$\text{Multi-head network (8 behaviors)} : 7,984 + 8 \times 577 = 12,600 \text{ parameters}$$

$$\text{Overhead of multi-head} : \frac{12,600 - 8,561}{8,561} = \frac{4,039}{8,561} = 47.2\%$$

353 .4 Architecture Visualization

354 .5 Training Results Across 30 Seeds

355 The multi-head architecture substantially outperforms the single-head baseline in training. Across
356 30 independent random seeds, the multi-head design achieves a mean final fitness of 0.927 ± 0.010

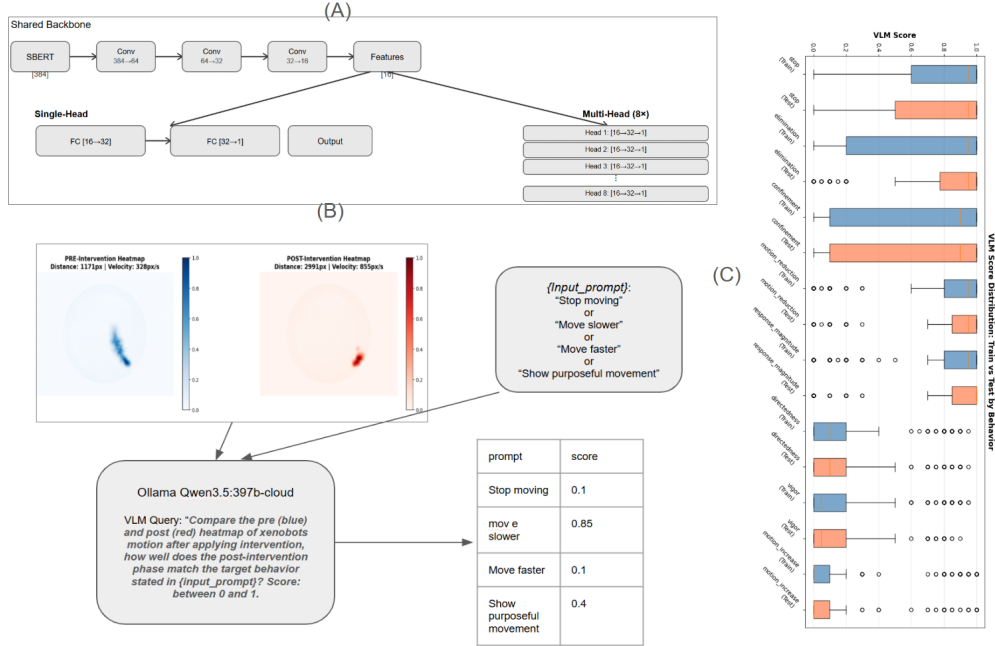


Figure 3: P2I network architecture and VLM reward scoring. (A) Single-head baseline: one output for all 8 behaviors causes gradient interference on opposing objectives. Multi-head design: 8 independent heads (577 parameters each) with shared backbone (7,984 parameters) enable specialization without interference. Parameter overhead: 47.2% (12,600 total for multi-head vs. 8,561 for single-head). (B) VLM reward scoring pipeline: pre/post motion heatmaps and prompt submitted to Qwen3.5-397B; VLM outputs score $s \in [0, 1]$ for behavior-prompt alignment. Example (batch-000070): velocity increases 2.6×, yielding high scores for motion-increase behaviors and low scores for opposing behaviors. (C) VLM score distribution by behavior, showing easy-tier concentration (0.63–0.81) and hard-tier sparsity (0.14–0.19), justifying stratified train-test split and difficulty-based analysis.

357 compared to single-head’s 0.666 ± 0.022 —a 39% absolute improvement (Wilcoxon $p = 7.54 \times 10^{-7}$,
358 indicating highly significant superiority).

359 This advantage arises from the architectural decoupling: each behavior’s head receives independent
360 learning signals without interference from opposing objectives. The single-head baseline, by contrast,
361 must compromise between conflicting goals (e.g., “stop” demands short stimulus duration while
362 “accelerate” demands long duration), leading to poor optimization on the shared output.

363 Per-behavior analysis reveals that hard-tier behaviors (directedness, vigor, motion_increase) benefit
364 most from the multi-head design. These sparse-reward behaviors achieve dramatic improvements:
365 directedness gains +0.60 (0.216 → 0.815), vigor +0.58 (0.254 → 0.833), and motion_increase +0.25
366 (0.674 → 0.926). Easy-tier behaviors, already well-represented in the archive, show smaller absolute
367 gains but consistently stable performance. This pattern reflects the regularization effect of multi-task
368 learning: shared backbone features learned from easy behaviors help bootstrap learning on hard
369 behaviors, preventing individual heads from overfitting to noise.

370 Figure 4 visualizes these training dynamics across all 100 CMA-ES generations.

371 Appendix B: CMA-ES Hyperparameters and Configuration

372 **CMA-ES Variant:** We use the *separable CMA-ES* (sep-CMA-ES) variant, which maintains a
373 diagonal covariance matrix instead of full-rank. This reduces memory complexity from $O(n^2)$ to
374 $O(n)$ and computational complexity from $O(n^3)$ to $O(n)$, making large-scale optimization tractable.
375 For $n = 12,600$ parameters, sep-CMA-ES requires ~50 MB memory (vs. ~27 GB for full-rank
376 CMA-ES), and scales linearly with parameter count.

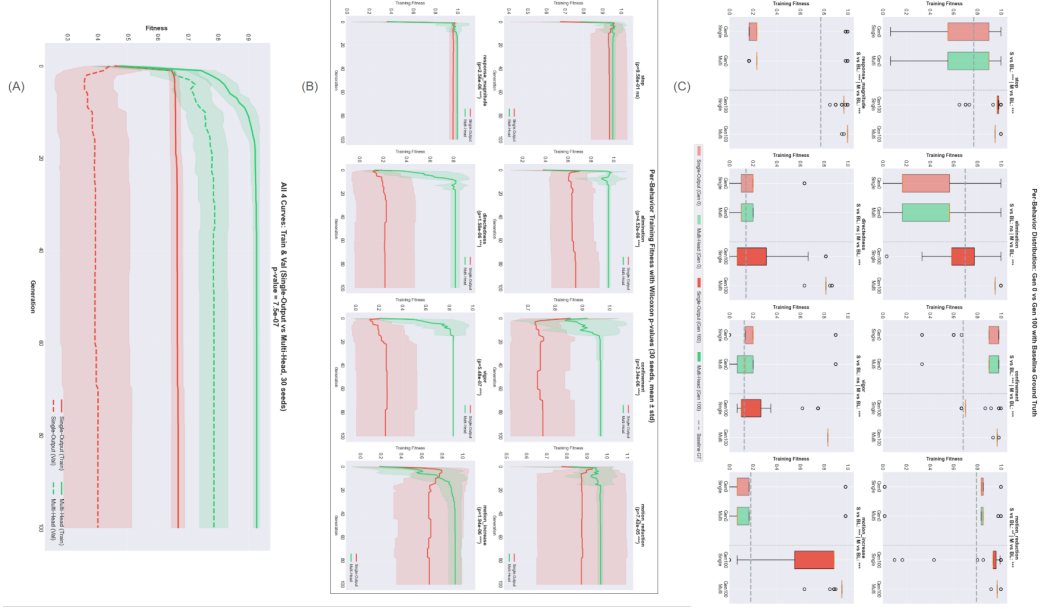


Figure 4: Training results: (A) Overall training fitness curves comparing single-head (red dashed) vs. multi-head (green solid) architecture across 100 CMA-ES generations. Multi-head achieves 0.927 ± 0.010 final fitness vs. single-head 0.666 ± 0.022 (Wilcoxon $p = 7.54e-07$). (B) Per-behavior training fitness evolution for multi-head architecture, showing independent optimization trajectories for all 8 behaviors without gradient interference. (C) Boxplot of final per-behavior performance, highlighting dramatic improvements on hard behaviors (directedness +0.60, vigor +0.58, motion_increase +0.25) while maintaining performance on easy behaviors.

Population size: $\lambda = 32$ (recommended: $\lambda = 4 + \lfloor 3 \ln(n) \rfloor = 4 + \lfloor 3 \times 9.44 \rfloor = 32$ with $n = 12,600$ total parameters; matches our actual choice)

Initial mutation strength: $\sigma_0 = 0.3$

Number of generations: 100 (training stops when fitness plateau is detected)

Combined fitness function:

$$\mathcal{F}_{\text{combined}} = \frac{1}{8} \sum_{i=1}^8 \mathcal{F}_i(\mathbf{w}) \quad (10)$$

Each behavior’s fitness is the mean VLM score across all training prompts assigned to that behavior.

Appendix C: Statistical Methods and Baseline Computation

.6 Wilcoxon Signed-Rank Test

We use Wilcoxon signed-rank test (one-sided alternative: multi-head $>$ single-head) to compare final fitness across 30 seeds. This non-parametric test is appropriate for non-normally distributed fitness distributions.

$$H_0 : \text{Multi-head fitness} = \text{Single-head fitness}$$

$$H_1 : \text{Multi-head fitness} > \text{Single-head fitness}$$

Significance level: $\alpha = 0.05$ (with Bonferroni correction for multiple behaviors: $\alpha/8 = 0.00625$)

.7 Effect Size

Cohen’s d is computed as:

$$d = \frac{\mu_{\text{multi}} - \mu_{\text{single}}}{\sqrt{\frac{(n-1)s_{\text{multi}}^2 + (n-1)s_{\text{single}}^2}{2(n-1)}}} \quad (11)$$

where $n = 30$ seeds, μ is mean, s is standard deviation.

.8 Baseline Computation

Dataset baselines represent the expected VLM score from random batch selection without model guidance. For each behavior i :

$$\text{Baseline}_i = \frac{1}{N_{\text{prompts}} \times N_{\text{batches}}} \sum_{p \in P_i} \sum_{b \in B} S(b, p, i) \quad (12)$$

where $S(b, p, i)$ is the VLM score for batch b , prompt p , and behavior i ; P_i is the set of training prompts for behavior i ; B is the set of training batches.

This gives per-behavior baseline scores representing the inherent difficulty: easy behaviors (high scores) have many good batches; hard behaviors (low scores) have few good batches.

Appendix D: Generalization Test Details

.9 GT1: Generalization to Novel Prompts

Test prompts: 5 held-out prompts per behavior (40 total, never seen during training)

Test batches: 99 training batches (same as training data)

Evaluation: For each of 40 test prompts, compute VLM score using the trained P2I network. Average score per behavior.

.10 GT2: Generalization to Novel Batches

Test prompts: Same 5 held-out prompts per behavior

Test batches: 44 held-out test batches (never seen during training)

Evaluation: For each of 40 test prompts $\times$ 44 test batches combination, compute VLM score. Average score per behavior.

Appendix E: Prompt Lists

.11 Training Prompts (40 total)

stop: "stop moving", "remain stationary", "become immobile",
"freeze in place", "cease all motion"

elimination: "make the bot disappeared", "bot ceases all activity",
"complete motion shutdown", "bot becomes unresponsive",
"permanent immobilization"

confinement: "stay confined to center", "localized activity",
"restricted mobility", "confined movement",
"center-seeking behavior"

motion_reduction: "move slower", "reduce movement", "minimize motion",
"decrease speed", "low velocity movement"

425
 426 response_magnitude: "dramatic behavior change", "pronounced effect of intervention",
 427 "stark difference from baseline", "strong behavior shift",
 428 "large response to intervention"
 429
 430 directedness: "move in directed paths", "show purposeful movement",
 431 "goal-seeking behavior", "clear trajectory intent",
 432 "directed exploration"
 433
 434 vigor: "show vigor", "vigorous movement", "energetic behavior",
 435 "powerful movement", "high energy movement"
 436
 437 motion_increase: "move faster", "increase activity level",
 438 "become more active", "high velocity movement",
 439 "increase speed"

440 .12 Test Prompts (40 total, novel variants)

441 stop: "remain stationary", "become immobile", "freeze in place",
 442 "halt movement", "stop all motion"
 443
 444 elimination: "total cessation of movement", "bot viability lost",
 445 "lethal intervention effect", "irreversible behavioral change",
 446 "bot elimination"
 447
 448 confinement: "center-restricted movement", "confined activity",
 449 "limited mobility range", "centralized behavior",
 450 "avoid periphery"
 451
 452 motion_reduction: "sluggish motion", "lethargic behavior",
 453 "minimize activity", "slower pace", "reduced speed"
 454
 455 response_magnitude: "substantial behavior shift", "marked behavioral change",
 456 "pronounced difference from baseline", "significant effect",
 457 "strong intervention response"
 458
 459 directedness: "intentional movement", "straight-line trajectory",
 460 "focused motion", "deliberate paths", "oriented behavior"
 461
 462 vigor: "forceful motion", "lively behavior", "robust movement",
 463 "intense activity", "spirited motion"
 464
 465 motion_increase: "accelerate motion", "boost activity", "rapid movement",
 466 "increased velocity", "amp up speed"

467 Appendix A.1: Multi-Task Learning Analysis

468 .13 Single-Task Experts vs Multi-Head Architecture

469 To directly test whether shared backbone improves generalization, we trained 8 independent expert
 470 networks (one per behavior) using CMA-ES, matching the multi-head training budget and protocol.
 471 Single-task experts achieved high training fitness (0.95 ± 0.04 mean), yet failed catastrophically on
 472 generalization: GT1 score drops to 0.5597 ± 0.3799 (35% degradation), with hard behaviors reaching
 473 only 8–30% of their training performance. Multi-head architecture achieves consistent GT1 across
 474 all behaviors (0.73–0.98), demonstrating superior generalization.

475 This pattern – perfect single-task training but poor generalization – aligns with multi-task learning
 476 theory (Caruana et al. [?]), where shared backbone representations enable hard tasks to benefit from
 477 learning easier tasks, thereby preventing overfitting.

Table 4: Single-task expert networks vs multi-head CMA-ES: per-behavior GT1 generalization. Single-task networks achieve high training fitness but severe overfitting, especially on hard behaviors. Multi-head achieves balanced generalization across all 8 behaviors.

Behavior	Single-Task Train	Single-Task GT1	Multi-Head GT1	Gain
<i>Easy-tier</i>				
stop	1.000	0.944	0.949	+0.005
motion_reduction	1.000	0.936	0.956	+0.020
response_magnitude	1.000	0.723	0.980	+0.257
elimination	1.000	0.731	0.942	+0.211
<i>Hard-tier</i>				
motion_increase	0.958	0.081	0.731	+0.650
vigor	0.882	0.139	0.737	+0.598
directedness	0.850	0.298	0.732	+0.434
confinement	0.979	0.627	0.823	+0.196
Mean	0.954	0.560	0.844	+0.284

478 .14 Mechanisms of Multi-Task Learning Success

479 Following Caruana et al., we identify three mechanisms enabling multi-head success:

480 *Representation bias toward shared minima.* The shared backbone simultaneously optimizes 8
481 behaviors, biasing search toward representations useful to all tasks rather than optimal for any single
482 task. Single-task networks, trained in isolation, optimize separately and often converge to local
483 minima that overfit to task-specific noise. Hard behaviors (motion_increase, directedness, vigor)
484 suffer most, achieving only 8–30% generalization despite perfect training. Shared learning forces
485 discovery of robust features that work across behaviors.

486 *Eavesdropping: hard tasks learn from easy tasks.* Easy behaviors (stop, motion_reduction) provide
487 strong, stable gradient signals. Hard behaviors, with sparser reward landscapes, can “listen to” feature
488 learning from easy tasks through the shared backbone. For example, the clear motion signal from
489 stop learning helps bootstrap representations that enable directedness learning. Single-task networks
490 lose this auxiliary signal, leaving hard behaviors to learn from weak gradients alone.

491 *Automatic regularization via multi-objective balance.* Training 8 objectives simultaneously prevents
492 overfitting to spurious features in any single task. Multi-head’s slightly lower training fitness
493 (0.876 vs 0.954 single-task) reflects necessary compromises; yet this produces dramatically higher
494 generalization (0.844 GT1 vs 0.560 single-task), indicating better regularization. The network learns
495 to prioritize features generalizable across behaviors.

496 The data strongly suggest that for this class of problems—learning from sparse, noisy signals across
497 multiple related objectives—multi-task learning with shared backbone is not optional but essential.

498 Appendix F: Per-Behavior Generalization Statistics (GT1 and GT2)

499 This appendix provides detailed per-behavior breakdowns of generalization test performance.

500 .15 Generalization to Novel Prompts (GT1)

501 All eight behaviors clear the $p < 10^{-7}$ threshold with Cohen’s d ranging from 2.32 to 42.15,
502 indicating large effect sizes. Variance explained (R^2) ranges from 0.84 to 0.999, indicating that the
503 learned mapping accounts for most score variation across novel prompts.

504 .16 Generalization to Novel Batches (GT2)

505 Six of eight behaviors show significant improvement over baseline ($p < 10^{-5}$). Two behaviors
506 — confinement and motion_reduction — show statistically significant negative transfer ($p <$
507 10^{-4}), likely due to distribution shift in test cohort duration statistics. The antagonistic behaviors

Table 5: GT1 per-behavior statistics: novel test prompts paired with training-set videos. Computed across 30 seeds; p -values from Wilcoxon signed-rank tests against the random-retrieval baseline.

Behavior	Baseline	GT1 mean	Improvement	p -value	Cohen’s d	R^2
stop	0.7496	0.9517	+27.0%	6.80×10^{-8}	22.52	0.998
motion_reduction	0.7633	0.9030	+18.3%	2.09×10^{-7}	2.58	0.870
motion_increase	0.1397	0.7340	+425.3%	7.05×10^{-7}	9.23	0.988
vigor	0.1559	0.7410	+375.3%	4.17×10^{-7}	20.00	0.998
response_magnitude	0.7623	0.9800	+28.6%	1.01×10^{-7}	42.15	0.999
directedness	0.1502	0.7323	+387.6%	2.61×10^{-7}	11.82	0.993
confinement	0.6309	0.9443	+49.7%	4.73×10^{-7}	2.32	0.843
elimination	0.6995	0.9383	+34.1%	3.32×10^{-7}	6.89	0.979

Table 6: GT2 per-behavior statistics: novel test prompts paired with held-out test videos. Computed across 30 seeds; p -values from Wilcoxon signed-rank tests against the random-retrieval baseline. Negative improvements indicate scores below baseline.

Behavior	Baseline	GT2 mean	Improvement	p -value	Cohen’s d	R^2
stop	0.7496	0.8717	+16.3%	8.66×10^{-3}	0.37	0.123
motion_reduction	0.7633	0.5580	−26.9%	6.34×10^{-7}	−1.08	0.537
motion_increase	0.1397	0.3993	+185.8%	9.74×10^{-7}	1.44	0.675
vigor	0.1559	0.6337	+306.5%	4.17×10^{-7}	4.97	0.961
response_magnitude	0.7623	0.9783	+28.3%	1.30×10^{-7}	2.48	0.860
directedness	0.1502	0.3420	+127.7%	6.02×10^{-7}	2.77	0.885
confinement	0.6309	0.2960	−53.1%	9.87×10^{-5}	−0.87	0.433
elimination	0.6995	0.9357	+33.8%	7.49×10^{-7}	2.25	0.835

(vigor, motion_increase, directedness) maintain substantial gains (+127% to +307%) even under the harder GT2 protocol.

17 Correlation Analysis: GT1 vs GT2

Pearson correlation between GT1 and GT2 performance per behavior:

Table 7: Pearson correlation: GT1 vs GT2 fitness across 30 seeds.

Behavior	Pearson r	p -value
stop	0.067	0.724
elimination	0.029	0.879
confinement	−0.487	0.006 **
motion_reduction	0.094	0.620
response_magnitude	0.370	0.044 *
directedness	0.660	< 0.001 ***
vigor	−0.391	0.033 *
motion_increase	−0.149	0.433

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Weak correlations suggest that GT1 and GT2 measure different generalization modes: semantic generalization (novel language) vs. biological transfer (novel batches). Some behaviors (directedness, response_magnitude) show positive correlation (good GT1 → good GT2), while others show negative correlation (good GT1 but poor GT2 on novel batches), indicating archive-limited behaviors.

516 Appendix G: REINFORCE vs CMA-ES Ablation Study

517 We compared CMA-ES with a policy gradient baseline (REINFORCE with baseline) to understand
 518 whether the superior CMA-ES performance reflects the choice of optimization algorithm or simply
 519 the need for more data/computation. This section documents the ablation.

520 .18 Why REINFORCE for Contextual Bandits?

521 Our problem is a contextual bandit: the environment provides a state (prompt embedding), the agent
 522 selects an action (intervention duration), and receives a scalar reward (VLM score) with no state
 523 transition. REINFORCE [?] is theoretically justified for this setting because:

- 524 • **Terminal rewards only:** No trajectory to optimize; single sample per episode suffices.
- 525 • **Fixed state representation:** Prompts are embedded to fixed 384-dim vectors; the policy
 526 learns a deterministic mapping.
- 527 • **Single action and reward:** No need for temporal credit assignment; advantage estimation
 528 reduces to reward minus baseline.
- 529 • **Sparse, noisy rewards:** Running-mean baseline [?] provides automatic variance reduction
 530 without learning a separate value network.

531 .19 REINFORCE Policy Gradient Formulation

532 The policy π_θ maps prompt embedding $\mathbf{e} \in \mathbb{R}^{384}$ and behavior b to a scalar action $a \in [0, 1]$
 533 representing predicted intervention duration:

$$a = \pi_\theta(\mathbf{e}, b) = \sigma(\text{MLP}(\mathbf{e}; \Theta_b)) \quad (13)$$

534 where σ is sigmoid and Θ_b are behavior-specific parameters.

535 The policy gradient objective is derived from the REINFORCE algorithm [?]:

$$\nabla_\theta J(\theta) = \mathbb{E}_{b, \mathbf{e}}[(G_t - V_b) \nabla_\theta \log \pi_\theta(a | \mathbf{e}, b)] \quad (14)$$

536 where $G_t = r$ is the terminal VLM reward and $V_b = \text{RunningMean}(R_{\text{history}}(b))$ is a per-behavior
 537 baseline for variance reduction. The running mean is updated incrementally:

$$V_b^{(k+1)} = \frac{k \cdot V_b^{(k)} + r_k}{k + 1} \quad (15)$$

538 The training loss includes entropy regularization to encourage exploration:

$$\mathcal{L}(\theta) = -\mathbb{E}[(r - V_b) \log \pi_\theta(a | \mathbf{e}, b)] + \beta H(\pi_\theta) \quad (16)$$

539 where $\beta = 0.01$ and $H(\pi_\theta) = -\sum_a \pi_\theta(a | \mathbf{e}, b) \log \pi_\theta(a | \mathbf{e}, b)$ is the policy entropy.

540 .20 REINFORCE Algorithm

541 .21 Training Configuration

542 REINFORCE training was configured to match CMA-ES' computational budget:

543 .22 Computational Budget Alignment

544 Both methods evaluate the same number of (behavior, prompt) pairs. CMA-ES performs:
 100 generations $\times$ 32 population $\times$ 8 behaviors $\times$ 3 prompts/behavior = 76,800 VLM evaluations
 (17)

545 REINFORCE samples behavior-prompt pairs uniformly during training:
 76,800 episodes $\times$ 1 (behavior, prompt) pair/episode = 76,800 VLM evaluations (18)

546 Thus both methods have equal computational cost in terms of VLM calls, ensuring a fair comparison.

Algorithm 1 REINFORCE for P2I Network Training

```

1: Input: Network  $\pi_\theta$ , learning rate  $\alpha = 10^{-3}$ , entropy weight  $\beta = 0.01$ 
2: Initialize: Random weights  $\theta_0$ , baseline history  $H_b = \{\}$  for each behavior  $b \in \{1, \dots, 8\}$ 
3: for episode  $k = 1, \dots, 76800$  do
4:   Sample behavior:  $b \sim \text{Uniform}(\{1, \dots, 8\})$ 
5:   Sample prompt:  $p \sim \text{Dataset}(b)$ 
6:   Encode:  $\mathbf{e} = \text{SentenceBERT}(p) \in \mathbb{R}^{384}$ 
7:   Forward pass:  $a = \pi_\theta(\mathbf{e}, b)$  (scalar action in  $[0, 1]$ )
8:   Map to duration:  $d_{\text{pred}} = d_{\text{min}} + a \cdot (d_{\text{max}} - d_{\text{min}})$ 
9:   Find nearest batch:  $b_{\text{near}} = \arg \min_j |d_j - d_{\text{pred}}|$ 
10:  Retrieve reward:  $r = \text{VLM}(b, p, b_{\text{near}})$  from precomputed scores
11:  Update baseline:  $H_b \leftarrow H_b \cup \{r\}$ 
12:  Compute advantage:  $A = r - \text{Mean}(H_b)$ 
13:  Compute loss:  $\mathcal{L} = -A \log \pi_\theta(a | \mathbf{e}, b) + \beta H(\pi_\theta(\cdot | \mathbf{e}, b))$ 
14:  Gradient step:  $\theta \leftarrow \theta + \alpha \nabla_\theta \mathcal{L}$ 
15: end for
16: Output: Trained policy  $\pi_\theta$ 

```

Table 8: REINFORCE vs CMA-ES computational alignment.

Metric	REINFORCE	CMA-ES
Episodes / Generations	76,800	100
Network evaluations per step	1	32
Behavior-prompt pairs per evaluation	1	$8 \times 3 = 24$
Total behavior-prompt pairs	76,800	76,800
Learning rate	0.001	—
Entropy weight	0.01	—
Baseline	Running mean/behavior	—

.23 Per-Behavior Results

Table 9 shows per-behavior training fitness across CMA-ES single-head, CMA-ES multi-head, and REINFORCE:

Table 9: Per-behavior training fitness: CMA-ES vs REINFORCE. CMA-ES: final generation, mean across 30 seeds. REINFORCE: final epoch, single run. Baseline VLM scores are computed on untrained random network.

Behavior	Baseline	Single-Head	Multi-Head	REINFORCE	CMA-ES/RF Ratio
stop	0.748	0.937	0.955	0.634	1.51
motion_reduction	0.763	0.872	0.967	0.809	1.19
motion_increase	0.140	0.674	0.926	0.338	2.74
vigor	0.156	0.254	0.833	0.285	2.92
response_magnitude	0.768	0.958	0.997	0.820	1.22
directedness	0.148	0.216	0.815	0.190	4.29
confinement	0.648	0.756	0.967	0.640	1.51
elimination	0.702	0.666	0.955	0.545	1.75

.24 Analysis: CMA-ES vs REINFORCE Performance and Mechanisms

Empirical Performance Disparities. Hard-tier behaviors reveal the largest gaps between methods. REINFORCE fails catastrophically on sparse-reward problems:

- Directedness: REINFORCE 0.190 vs CMA-ES 0.815 ($4.3 \times$ gap)
- Vigor: REINFORCE 0.285 vs CMA-ES 0.833 ($2.9 \times$ gap)

555 • Motion_increase: REINFORCE 0.338 vs CMA-ES 0.926 ($2.7\times$ gap)

556 The policy gradient method barely improves over baseline (only +28% on directedness), while
 557 CMA-ES achieves +452%. Easy behaviors show mixed results; REINFORCE occasionally matches
 558 CMA-ES on response_magnitude (0.820 vs 0.997, $1.22\times$ gap) and motion_reduction (0.809 vs 0.967,
 559 $1.19\times$ gap), but consistently underperforms. Averaging across all behaviors: REINFORCE 0.533 vs
 560 CMA-ES multi-head 0.927 (73% improvement).

561 **Why REINFORCE Fails: Three Mechanisms.** The policy gradient method struggles for fundamen-
 562 tal reasons:

- 563 1. **High variance on sparse rewards.** REINFORCE uses single-trajectory gradient estimates
 564 with variance scaling as $O(T^2)$ in episode length. With only 9,600 samples per behavior
 565 (76,800 episodes $\div$ 8 behaviors), the gradient signal is drowned by noise. Even with
 566 baseline subtraction, variance remains $O(T)$, insufficient for convergence on behaviors
 567 where $P(\text{good intervention}) \approx 0.3$.
- 568 2. **Poor credit assignment on multi-head network.** The network has 8 independent output
 569 heads, one per behavior. During training on behavior b_j , only head j receives learning
 570 signals. The shared backbone simultaneously receives conflicting gradients from all 8
 571 behaviors, each pulling the shared weights toward different optima. CMA-ES avoids this by
 572 optimizing the entire weight vector as a unit, naturally balancing multi-objective pressure.
- 573 3. **Unreliable gradient estimation on discrete behavior selection.** Although the network
 574 outputs a continuous duration value, it implicitly selects a behavior during sampling. The
 575 gradient estimate $\nabla_{\theta} \log \pi_{\theta}(a|e, b) \cdot (r - V_b)$ is biased when the discrete behavior choice
 576 has not been explored sufficiently; baseline subtraction cannot correct this discrete-action
 577 bias.

578 CMA-ES avoids these issues through population-based exploration and black-box optimization,
 579 making it more robust on sparse-reward, high-dimensional, multi-objective problems.

580 Appendix H: Video Processing and Prompt Routing

581 .25 Video Processing: Motion Denoising Pipeline

582 Raw xenobot videos contain optical noise (bubbles, electrical reflections) that confounds motion
 583 analysis. We denoise using a four-step pipeline:

- 584 1. **Frame differencing:** Consecutive frames are differenced to detect motion regions.
- 585 2. **HSV color thresholding:** White pixels (low saturation < 20 , corresponding to bubbles
 586 and reflections) are rejected; colored pixels (high saturation, corresponding to pigmented
 587 xenobots) are retained.
- 588 3. **Morphological operations:** Erosion and dilation clean isolated noise artifacts.
- 589 4. **Optical flow on masked regions:** Dense optical flow (Farneback method) is computed only
 590 on the bot mask, yielding motion magnitude heatmaps.

591 The motion heatmap is computed as:

$$\Delta H = \sum_t \text{Magnitude}_{\text{masked}}^{\text{post}}(t) - \sum_t \text{Magnitude}_{\text{masked}}^{\text{pre}}(t) \quad (19)$$

592 where magnitude is accumulated across all consecutive frame pairs in the pre- and post-intervention
 593 periods.

594 .26 Prompt Embeddings and Routing

595 Prompts are embedded using SentenceBERT (SBERT), a sentence transformer pretrained on natural
 596 language inference tasks. SBERT produces 384-dimensional embeddings that capture semantic
 597 similarity.

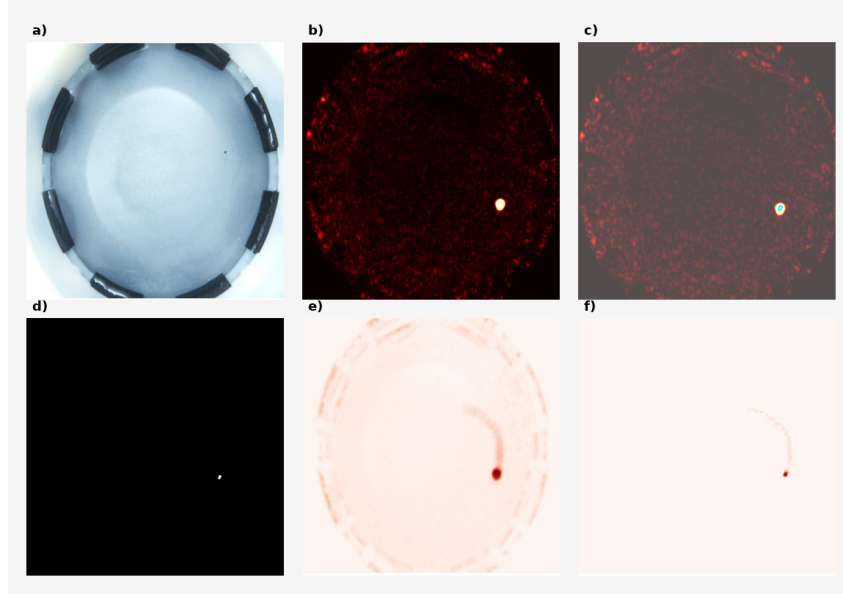


Figure 5: Motion denoising pipeline for xenobot behavioral analysis. (a) Raw video frame showing multi-electrode array chamber with xenobot. Bubbles (translucent white spheres) and electrical reflections create optical noise. (b) Optical flow magnitude heatmap computed from raw consecutive frames (frame t to $t + 5$). Red indicates motion regions; extensive noise from bubbles and reflections throughout the chamber confounds true xenobot motion signal. (c) Frame differencing mask: bright regions indicate pixels with intensity change between frames. Detects all motion (xenobot and bubbles). (d) HSV saturation mask: white pixels (low saturation < 20) corresponding to bubbles and reflections are rejected; colored pixels (high saturation) corresponding to pigmented xenobots are retained. (e) Combined bot mask: element-wise intersection of motion and color masks. Xenobot regions preserved; bubble noise eliminated through color filtering. (f) Cleaned optical flow magnitude heatmap: only motion in bot regions shown. This represents true xenobot behavior without artifact noise. Magnitude values accumulated across all consecutive frame pairs to compute ΔH , the net behavioral change between pre- and post-intervention periods.

.26.1 Training Routing (Oracle)

During training, prompts are assigned to their true behavior class (oracle routing). The CMA-ES optimizer then maximizes fitness for each behavior’s assigned head.

.26.2 Test Routing (Cosine Similarity)

At test time, the true behavior label is not available. Instead, we use cosine similarity routing: each test prompt is routed to the training prompt (from the 40 training prompts) with highest embedding similarity. The model then uses the corresponding behavior head.

This approach allows generalization to novel prompts without requiring the model to learn behavior classification; it leverages the assumption that semantically similar prompts should map to the same behavior head.

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Justification: All experiments run on CPU-only (Intel i9-14900KF, 24 cores, 128GB RAM; GPU not required for optimization). CMA-ES optimization uses a diagonal covariance matrix (separable CMA-ES variant, sep-CMA-ES) to maintain $O(n)$ memory complexity, making it feasible for the 12,600-parameter P2I network. Per-seed CMA-ES: 25–35 minutes (100 generations, population 32). Total: 30 seeds $\times$ 30 min = 15 hours wall-clock time. VLM precomputation (offline, one-time): Qwen3.5-397B on cloud API with 4 parallel workers to score 143 batches $\times$ 80 prompts $\times$ 8 behaviors (91,520 evaluations). Subsequent CMA-ES training is CPU-only, single-threaded. Total experimental reproducibility: CMA-ES 15 hours CPU time (no GPU required); VLM scoring time depends on cloud API throughput.

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